

## T3 group factors: EFT\_2\_after\_S1\_S2

Active theory: SMEFT with the Weinberg operator  $C_5$ . No T3 heavy field remains dynamical.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table,  $\mathcal{S}$  denotes the RGE structure multiplying the displayed group factor in  $16\pi^2\beta$ .

## Notation key

| symbol      | interaction / definition                                                      |
|-------------|-------------------------------------------------------------------------------|
| $g_1$       | $D_\mu \supset ig_1 Y B_\mu$                                                  |
| $g_2$       | $D_\mu \supset ig_2 T^A W_\mu^A$                                              |
| $g_3$       | $D_\mu \supset ig_3 t^A G_\mu^A$                                              |
| $Y_u$       | $-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$                                      |
| $Y_d$       | $-\bar{Q} Y_d H d + \text{h.c.}$                                              |
| $Y_e$       | $-\bar{L} Y_e H e + \text{h.c.}$                                              |
| $T$         | $\text{Tr}\left(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d\right)$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1(H^\dagger H)^2$                                         |
| $C_5^{ij}$  | $(L_i^T C L_j) H H$                                                           |

## Stage field content

| run  | $S_1$               | $S_2$              | $F$      | integrated at this stage | active after matching |
|------|---------------------|--------------------|----------|--------------------------|-----------------------|
| T3-B | $(2, -\frac{1}{2})$ | $(2, \frac{1}{2})$ | $(1, 0)$ | S1, S2                   | none                  |

## Decoupling statement

All dependence on the original  $S_1$ ,  $S_2$  and  $F$  representations is contained in the matched boundary value  $C_5(\mu_{\text{th}})$ . The one-loop running below the final threshold is common to all T3 classes.

$\beta_{C_5}$

| $\mathcal{S}$             | T3 – B ( $\alpha = -1$ ) |
|---------------------------|--------------------------|
| $\lambda_1 C_5$           | 2                        |
| $g_2^2 C_5$               | -3                       |
| $T C_5$                   | 2                        |
| $(Y_e^\dagger Y_e)^T C_5$ | $-\frac{3}{2}$           |
| $C_5(Y_e^\dagger Y_e)$    | $-\frac{3}{2}$           |

## Flavor form

$$16\pi^2 \frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \quad (1)$$

$$-\frac{3}{2} \left[ (Y_e^\dagger Y_e)^T C_5 + C_5 (Y_e^\dagger Y_e) \right]. \quad (2)$$